LATIHAN 7

A. selesaikan SPL dengan cramer

1. $x + 3y + 2z = 16$
   $2x + 4y - 2z = 12$
   $x + y + 4z = 20$
2. $2x_1 + x_2 + x_3 = 12$
   $x_1 + 2x_2 - x_3 = 3$
   $3x_1 - x_2 + x_3 = 11$

B. selesaikan SPL dengan matriks

3. $x + y - z = -3$
   $x + 2y + z = 7$
   $2x + y + z = 4$
4. $x + y + z = 3$
   $2x + y - 5z = -8$
   $3x - 2y + z = 5$

C. Diketahui SPL :

$2x_1 + 5x_2 - 3x_3 = 3$
$6x_1 + 8x_2 - 5x_3 = 7$
$-3x_1 + 3x_2 + 4x_3 = 15$

- Tuliskan SPL dalam bentuk matriks $AX = B$
- Hitung Determinan A dengan sorus
- Tentukan Adj (A) dan A invers

Jawaban

A. 1. $D = \begin{vmatrix} 1 & 3 & 2 \\ 2 & 4 & -2 \\ 1 & 1 & 4 \end{vmatrix} \begin{matrix} 1 & 3 \\ 2 & 4 \\ 1 & 1 \end{matrix}$

$= (1.4.4) + (3.(-2).1) + (2.2.1) - (3.2.4) (1.(-2).1) (2.4.1)$

$= (16 + (-6) + 4) - (24 + (-2) + 8) = 14 - 30 = -16.$

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$$D_x = \begin{vmatrix} 16 & 3 & 2 \\ 12 & 4 & -2 \\ 20 & 1 & 4 \end{vmatrix} \begin{matrix} 16 & 3 \\ 12 & 4 \\ 20 & 1 \end{matrix}$$

$= (16.4.4) + (3.(-2).20) + (2.12.1) - (3.12.4) + (16.(-2).1) + (2.4.20)$

$= (256 + (-120) + 24) - (144 + (-32) + 160) = 160 - 272 = -112$

$$D_y = \begin{vmatrix} 1 & 16 & 2 \\ 2 & 12 & -2 \\ 1 & 20 & 4 \end{vmatrix} \begin{matrix} 1 & 16 \\ 2 & 12 \\ 1 & 20 \end{matrix}$$

$= (1.12.4) + (16.(-2).1) + (2.2.20) - (16.2.4) + (1.(-2).(20) + (2.12.1)$

$= (48 + (-32) + 80) - (128 + (-40) + 24) = 96 - 112 = -16$

$$D_z = \begin{vmatrix} 1 & 3 & 16 \\ 2 & 4 & 12 \\ 1 & 1 & 20 \end{vmatrix} \begin{matrix} 1 & 3 \\ 2 & 4 \\ 1 & 1 \end{matrix}$$

$= (1.4.20) + (3.12.1) + (16.2.1) - (3.2.20) + (1.12.1) + (16.4.1)$

$= (80 + 36 + 32) - (120 + 12 + 64) = 148 - 196 = -48$

$x = \frac{D_x}{D} = \frac{-112}{-16} = 7$ $\quad z = \frac{D_z}{D} = \frac{-48}{-16} = 3$

$y = \frac{D_y}{D} = \frac{-16}{-16} = 1$

2. $$D = \begin{vmatrix} 2 & 1 & 1 \\ 1 & 2 & -1 \\ 3 & -1 & 1 \end{vmatrix} \begin{matrix} 2 & 1 \\ 1 & 2 \\ 3 & -1 \end{matrix}$$

$= (2.2.1) + (1.(-1).3) + (1.1.(-1) - (1.1.1) + (2.(-1).(-1) + (1.2.3)$

$= ((4 + (-3) + (-1)) - (1 + 2 + 6) = 0 - 9 = -9$

$$D_x = \begin{vmatrix} 12 & 1 & 1 \\ 3 & 2 & -1 \\ 11 & -1 & 1 \end{vmatrix} \begin{matrix} 12 & 1 \\ 3 & 2 \\ 11 & -1 \end{matrix}$$

$= (12.2.1) + (1.(-1).11) + (1.3.(-1) - (1.3.1) + (12.(-1).(-1) + (1.2.11)$

$= ((24 + (-11) + (-3)) - (3 + 12 + 22) = 10 - 37 = -27$

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$$Dy = \begin{vmatrix} 2 & 12 & 1 \\ 1 & 3 & -1 \\ 3 & 11 & 1 \end{vmatrix} \begin{matrix} 2 & 12 \\ 1 & 3 \\ 3 & 11 \end{matrix}$$

$= (2.3.1) + (12.(-1).3) + (1.1.11) - (12.1.1) + (2.(-1).11) + (1.3.3)$

$= (6 + (-36) + 11) - (12 + (-22) + 9) = (-19) - (-1) = -18$

$$Dz = \begin{vmatrix} 2 & 1 & 12 \\ 1 & 2 & 3 \\ 3 & -1 & 11 \end{vmatrix} \begin{matrix} 2 & 1 \\ 1 & 2 \\ 3 & -1 \end{matrix}$$

$= (2.2.11) + (1.3.3) + (12.1.(-1) - (1.1.11) + (2.3.(-1) + (12.2.3)$

$= (44 + 9 + (-12) - (11 + (-6) + 72) = 41 - 77 = -36$

$x = \frac{Dx}{D} = \frac{-27}{-9} = 3$ $\qquad z = \frac{Dz}{D} = \frac{-36}{-9} = 4$

$y = \frac{Dy}{D} = \frac{-18}{-9} = 2$

B. 3. $x + y - 2 = -3$

$x + 2y + 2 = 7$ $\longrightarrow$ $\begin{pmatrix} 1 & 1 & -1 \\ 1 & 2 & 1 \\ 2 & 1 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} -3 \\ 7 \\ 4 \end{pmatrix}$

$2x + y + 2 = 4$

$$\text{Det}(A) = \begin{vmatrix} 1 & 1 & -1 \\ 1 & 2 & 1 \\ 2 & 1 & 1 \end{vmatrix}$$

$$= 1 \begin{vmatrix} 2 & 1 \\ 1 & 1 \end{vmatrix} - 1 \begin{vmatrix} 1 & 1 \\ 2 & 1 \end{vmatrix} + (-1) \begin{vmatrix} 1 & 2 \\ 2 & 1 \end{vmatrix}$$

$= 1(2 - 1) - 1(1 - 2) + (-1)(1 - 4)$

$= 1 - (-1) + 3 = 5$

- cari $A^{-1} = \frac{1}{\det A} \cdot \text{Adj } A \rightsquigarrow (\text{kof } A)^T$
- kofaktor:

$A_{11} = (-1)^2 \begin{vmatrix} 2 & 1 \\ 1 & 1 \end{vmatrix} = 1(2 - 1) = 1$

$A_{23} = (-1)^5 \begin{vmatrix} 1 & 1 \\ 2 & 1 \end{vmatrix} = -1(1 - 2) = 1$

$A_{12} = (-1)^3 \begin{vmatrix} 1 & 1 \\ 2 & 1 \end{vmatrix} = -1\,(1-2) = 1$

$A_{13} = (-1)^4 \begin{vmatrix} 1 & 2 \\ 2 & 1 \end{vmatrix} = 1\,(1-4) = -3$

$A_{21} = (-1)^3 \begin{vmatrix} 1 & -1 \\ 1 & 1 \end{vmatrix} = -1\,(1-(-1) = -2$

$A_{22} = (-1)^4 \begin{vmatrix} 1 & -1 \\ 2 & 1 \end{vmatrix} = 1\,(1-(-2) = 3$

$A_{31} = (-1)^4 \begin{vmatrix} 1 & -1 \\ 2 & 1 \end{vmatrix} = 1\,(1-(-2) = 3$

$A_{32} = (-1)^5 \begin{vmatrix} 1 & -1 \\ 1 & 1 \end{vmatrix} = -1\,(1-(-1) = -2$

$A_{33} = (-1)^6 \begin{vmatrix} 1 & 1 \\ 1 & 2 \end{vmatrix} = 1\,(2-1) = 1$

$$\begin{bmatrix} 1 & 1 & -3 \\ -2 & 3 & 1 \\ 3 & -2 & 1 \end{bmatrix} = \begin{bmatrix} 1 & -2 & 3 \\ 1 & 3 & -2 \\ -3 & 1 & 1 \end{bmatrix}^T$$

$$\frac{1}{5} \begin{pmatrix} 1 & -2 & 3 \\ 1 & 3 & -2 \\ -3 & 1 & 1 \end{pmatrix} \begin{pmatrix} -3 \\ 7 \\ 4 \end{pmatrix}$$

$$= \frac{1}{5} \begin{pmatrix} -3 & -14 & 12 \\ -3 & 21 & -8 \\ 9 & 7 & 4 \end{pmatrix}$$

$$\rightarrow \frac{1}{5} \begin{pmatrix} -5 \\ 10 \\ 20 \end{pmatrix} = \begin{pmatrix} -1 \\ 2 \\ 4 \end{pmatrix}$$

4. $x + y + z = 3$

$2x + y - 5z = -8$

$3x - 2y + z = 5$

$$\text{Det}(A) = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 1 & -5 \\ 3 & -2 & 1 \end{bmatrix} \begin{matrix} 1 & 1 \\ 2 & 1 \\ 3 & -2 \end{matrix}$$

$= (1.1.1) + (1.(-5).3) + (1.2.(-2) - [1.2.1) + (1.(-5).(-2) + (1.1.3)$

$= [1 + (-15) + (-4)] - (2 + 10 + 3) = (-18) - 15 = -33$

$A_{11} = (-1)^2 \begin{vmatrix} 1 & -5 \\ -2 & 1 \end{vmatrix} = 1\,(1-10) = -9$

$A_{12} = (-1)^3 \begin{vmatrix} 2 & -5 \\ 3 & 1 \end{vmatrix} = -1\,(2-(-15) = -17$

$A_{13} = (-1)^4 \begin{vmatrix} 2 & 1 \\ 3 & -2 \end{vmatrix} = -7$

$A_{21} = (-1)^3 \begin{vmatrix} 1 & 1 \\ -2 & 1 \end{vmatrix} = -1\,(1-(-2) = -3$

$A_{22} = (-1)^4 \begin{vmatrix} 1 & 1 \\ 3 & 1 \end{vmatrix} = 1\,(1-3) = -2$

$A_{23} = (-1)^5 \begin{vmatrix} 1 & 1 \\ 3 & -2 \end{vmatrix} = 5$

$A_{31} = (-1)^4 \begin{vmatrix} 1 & 1 \\ 1 & -5 \end{vmatrix} = 1(-5-1) = -6$

$A_{32} = (-1)^5 \begin{vmatrix} 1 & 1 \\ 2 & -5 \end{vmatrix} = -1(-5-2) = 7$

$A_{33} = (-1)^6 \begin{vmatrix} 1 & 1 \\ 2 & 1 \end{vmatrix} = 1(1-2) = -1$

$$= \begin{bmatrix} -9 & -17 & -7 \\ -3 & -2 & 5 \\ -6 & 7 & -1 \end{bmatrix} = \begin{bmatrix} -9 & -3 & -6 \\ -17 & -2 & 7 \\ -7 & 5 & -1 \end{bmatrix}^T$$

$$= \frac{1}{-33} \begin{pmatrix} -9 & -3 & -6 \\ -17 & -2 & 7 \\ -7 & 5 & -1 \end{pmatrix}^T \begin{pmatrix} 3 \\ -8 \\ 5 \end{pmatrix}$$

$$= \frac{1}{-33} \begin{pmatrix} -27 & 24 & -30 \\ -51 & 16 & 35 \\ -21 & -40 & -5 \end{pmatrix}$$

$$= \frac{1}{-33} \begin{pmatrix} -33 \\ 0 \\ -66 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix}$$

C.

$2x_1 + 5x_2 - 3x_3 = 3$

$6x_1 + 8x_2 - 5x_3 = 7$

$-3x_1 + 3x_2 + 4x_3 = 15$

- Tuliskan SPL dalam bentuk matriks $AX = B$

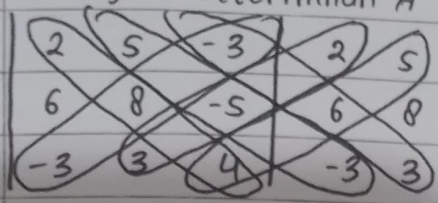
$$\begin{pmatrix} 2 & 5 & -3 \\ 6 & 8 & -5 \\ -3 & 3 & 4 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 3 \\ 7 \\ 15 \end{pmatrix}$$

$A \quad X = B$

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- Hitunglah determinan A dengan sarus



$(64 + 75 + (-54)) - (120 + (-30) + 72) = 85 - 162 = -77$

- Tentukan Adj (A) dan A invers

$$\text{Det } A = \begin{vmatrix} 2 & 5 & -3 \\ 6 & 8 & -5 \\ -3 & 3 & 4 \end{vmatrix} \begin{matrix} 2 & 5 \\ 6 & 8 \\ -3 & 3 \end{matrix} = -77$$

$A_{11} = (-1)^2 \begin{vmatrix} 8 & -5 \\ 3 & 4 \end{vmatrix} = 1\,(32 - (-15)) = 47$

$A_{12} = (-1)^3 \begin{vmatrix} 6 & -5 \\ -3 & 4 \end{vmatrix} = -1\,(24 - 15) = -9$

$A_{13} = (-1)^4 \begin{vmatrix} 6 & 8 \\ -3 & 3 \end{vmatrix} = 1\,(18 - (-24)) = 42$

$A_{21} = (-1)^3 \begin{vmatrix} 5 & -3 \\ 3 & 4 \end{vmatrix} = -1\,(20 - (-9)) = -29$

$A_{22} = (-1)^4 \begin{vmatrix} 2 & -3 \\ -3 & 4 \end{vmatrix} = 1\,(8 - 9) = -1$

$A_{23} = (-1)^5 \begin{vmatrix} 2 & 5 \\ -3 & 3 \end{vmatrix} = -1\,(6 - (-15)) = -21$

$A_{31} = (-1)^4 \begin{vmatrix} 5 & -3 \\ 8 & -5 \end{vmatrix} = 1\,(-25 - (-24)) = -$

$A_{32} = (-1)^5 \begin{vmatrix} 2 & -3 \\ 6 & -5 \end{vmatrix} = -1\,(-10 - (-18)) = -8$

$A_{33} = (-1)^6 \begin{vmatrix} 2 & 5 \\ 6 & 8 \end{vmatrix} = 1\,(16 - 30) = -14$

$$\text{Adj } A = \begin{bmatrix} 47 & -29 & - \\ -9 & -1 & -8 \\ 42 & -21 & -14 \end{bmatrix}^T \Leftarrow \begin{bmatrix} 47 & -9 & 42 \\ -29 & -1 & -21 \\ - & -8 & -14 \end{bmatrix}$$

$$A^{-1} = \frac{1}{\det A} \text{Adj } A$$

$$= \frac{1}{-77} \begin{pmatrix} 47 & -29 & - \\ -9 & -1 & -8 \\ 42 & -21 & -14 \end{pmatrix} \begin{pmatrix} 3 \\ 7 \\ 15 \end{pmatrix}$$

$$= \frac{1}{-77} \begin{pmatrix} 141 & -203 & -15 \\ -27 & -7 & -120 \\ 126 & -147 & -210 \end{pmatrix} = \frac{1}{-77} \begin{pmatrix} -77 \\ -154 \\ -231 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$$